

TANMAY SHARMA

sharma.tanm@northeastern.edu | Boston, Massachusetts | tanmayspace.com | [GitHub](https://github.com/tanmaysharma) | [LinkedIn](https://www.linkedin.com/in/tanmaysharma)

EDUCATION

Northeastern University | GPA: 3.58/4.00

September 2024 - Present

Master of Science in Cybersecurity

Boston, MA

- Relevant Coursework: Network Security, Binary Exploitation, Reverse Engineering, CTF, Cyberlaw.

SRM Institute of Science and Technology | GPA: 3.59/4.00

September 2020 - July 2024

Bachelor of Technology in Computer Science and Engineering

Chennai, India

- Relevant Coursework: Operating Systems, Computer Architecture, Network Security and Offensive Security.

TECHNICAL SKILLS

- **Development:** C/C++, Python, Linux x86 Assembly(32bit/64bit), Bash, Powershell, Rust(learning), SQL
- **Tools:** Metasploit, Wireshark, GDB Debugger, Ghidra, Nikto
- **Native Hacking:** Stack & Heap Buffer Overflow, Format String Vulnerability, ROP, Ret2Libc Attack
- **Network Security:** cryptographic protocols, SSL/TLS, IPsec, OAuth, intrusion detection, vulnerability analysis, secure application development, penetration testing, cyber attack mitigation.
- **Pen Testing:** Attacks on Active Directory, OSINT, Windows and Linux Privilege Escalation
- **Misc:** Raspberry Pi, ESP32 Microcontrollers, Flipper Zero

PROJECTS

Secure Chat Application

February 2020 - April 2025

- Engineered a terminal-based secure chat system leveraging SRP for password-authenticated key exchange, ECDH for secure key derivation, and AES-GCM for end-to-end encrypted messaging, enabling confidential and authenticated real-time communication among users, developed as part of the CY6740 course - Network Security.
- Designed and implemented robust message rate limiting and online user listing features to enhance system integrity and user experience, reducing potential abuse vectors by 80%
- Developed a socket-based client-server-client architecture supporting encrypted command-based interactions, strengthening the platform's resilience against network-based attacks

Phishing Hook

October 2024

- Developed a phishing detection web application leveraging deep learning to analyze URLs and webpage content, achieving 92% detection accuracy and significantly reducing phishing risks for end users
- Built an interactive web interface enabling real-time URL analysis and instant threat feedback, empowering users to proactively defend against phishing attacks

PROFESSIONAL EXPERIENCE

CyberspACe securiTy and forensics lab (CactiLab)

March 2024 – Present

Member

Boston, MA

- Collaborated on reverse engineering the EdgeTPU driver on Pixel devices, analyzing its interaction with firmware, which enhanced the lab's understanding of hardware security layers.
- Analyzed FirmXRay software by OSUSecLab, focusing on baseaddresssolver.java module, to identify how base addresses are resolved and features extracted, improving vulnerability assessment capabilities in embedded firmware.

Institute for Plasma Research (IPR)

September 2023 – November 2023

Project Intern

Ahmedabad, India

- Designed and deployed a secure, cost-effective network of Raspberry Pi-based access points, creating a dedicated wireless-to-wireless VPN/Wi-Fi solution for an isolated laboratory of 25–30 plasma physicists, enabling seamless and secure intra-lab communication.
- Implemented advanced encryption and security protocols, reducing network vulnerability by 60% and ensuring compliance with stringent research data protection standards, critical for sensitive plasma research experiments.
- Optimized network performance through real-time monitoring and custom configuration, achieving 99% uptime and improving data accessibility for researchers, which enhanced collaboration and accelerated experimental workflows by 30%.

CERTIFICATIONS & ACHIEVEMENTS

- **Certifications:** NSE 1-8 certifications, Palo Alto fundamentals of network and cloud security
- **In Progress:** PNPT (Practical Network Penetration Tester)
- **Achievements:** SimSpace Cyber Cup 2025 Top 6, Pwn.college white belt